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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel International GCSE (9–1)

Time 1 hour 30 minutes

Paper reference **4IT1/01**

Information and Communication Technology (ICT)

PAPER 1: Written Paper

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over

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Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A family uses digital devices in their everyday lives.

(Source: © PAL)

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1 Howard buys a tablet computer.

(a) Give **two** types of wireless connectivity that the tablet computer could use.

(2)

1 .....

2 .....

(b) Give **two** types of peripheral storage media that the tablet computer could use.

(2)

1 .....

2 .....

(c) Which **one** of these is required for the tablet computer to connect to the Internet?

(1)

☒ **A** GPS

☒ **B** ISP

☒ **C** LAN

☒ **D** NFC

(d) Which **one** of these is found in all computers?

(1)

☒ **A** Control software

☒ **B** Keyboard

☒ **C** Operating system

☒ **D** Processor

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(e) Which **one** of these would be used as a desktop replacement computer?

(1)

- ☒ **A** Laptop
- ☒ **B** Mainframe
- ☒ **C** Media Player
- ☒ **D** Server

(f) Describe how tablet computers allow people to work from home.

(2)

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(g) A CPU has 16 MiB of cache memory.

Construct an expression to show how many bits are in 16 MiB.

(3)

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(h) Howard sells his old hard disk drive to Steve.

(i) Complete the table by ranking these storage devices in order of their read/write speed.

A) Hard disk drive
B) Optical disk drive
C) Solid state drive

(2)

1 (Fastest)	
2	
3 (Slowest)	

(ii) Describe **two** factors that affect the read/write speed of a hard disk drive.

(4)

1 .....

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2 .....

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(iii) Explain **one** benefit to the environment of buying previously-owned devices.

(2)

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(iv) Which **one** of these is the type of software Howard should use to make copies of his applications?

(1)

☒ **A** Control

☒ **B** Office-productivity

☒ **C** Project management

☒ **D** Utility

(Total for Question 1 = 21 marks)

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2 Judy is researching online.

(a) List **two** types of user-generated reference sites Judy could use.

(2)

1 .....

2 .....

(b) Judy uses a mobile phone to make a voice call.

Draw a diagram to show the mobile phone's input and output devices required to make the call.

Label each device you use in your diagram.

(3)

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(c) Judy sets file permissions on her online documents.

Which **one** of these is a benefit of doing this?

(1)

- ☒ **A** Improves encryption
- ☒ **B** Increases storage
- ☒ **C** Quicker backup
- ☒ **D** Secures data

(d) Describe how a CAPTCHA test reduces the function of malware.

(2)

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(e) Explain **one** positive impact of cloud-based services on learning.

(2)

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(f) Describe how mobile broadband can be used to provide access to cloud-based services.

(2)

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(g) Give **one** reason why Judy should use mobile broadband.

(1)

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(h) Explain **one** benefit of using wired connectivity.

(2)

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(Total for Question 2 = 15 marks)

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3 Eunice installs some software on her laptop.

(a) Draw **one** straight line from **each** example of utility software to the correct purpose.

(3)

Utility software	**Purpose**
Compression	Protect from viruses
Back-up	Create copies of files
Anti-malware	Reduce file sizes
	Block IP addresses
	Convert file types

(b) Which **one** of these types of device can have software installed to it?

(1)

- ☒ **A** Memory
- ☒ **B** OMR Reader
- ☒ **C** Processor
- ☒ **D** Storage

(c) Explain **one** benefit to Eunice of updating her software regularly.

(2)

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(d) Eunice is travelling by train.

She connects to the Internet using the train's Wi-Fi connection.

Explain why the Wi-Fi provider blocks video streaming.

(2)

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(e) The Internet has provided a wider range of entertainment and leisure opportunities.

One example is video and audio streaming.

State **two other** examples.

(2)

1 .....

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2 .....

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(f) Eunice changes her laptop's GPS settings.

Which **one** of these statements is true?

(1)

☒ **A** GPS is used by all digital devices

☒ **B** GPS does not use an Internet connection

☒ **C** GPS uses cloud computing

☒ **D** GPS is better if open source software is used

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(g) Eunice writes a blog.

(i) Eunice wants visitors to her blog to see posts that are likely to be relevant to them.

Describe how cookies can be used to achieve this.

(4)

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(ii) In some countries, laws require website owners to obtain 'informed consent' from visitors before using cookies.

Explain why these laws are required.

(2)

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(Total for Question 3 = 17 marks)

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4 Hugh sells his music online.

(a) Which **one** of these is a law that protects his work?

(1)

- ☒ **A** Computer Misuse Act
- ☒ **B** Copyright, Designs and Patents Act
- ☒ **C** Human Rights Act
- ☒ **D** Data Protection Act

(b) List **three** benefits to Hugh of selling his music online.

(3)

1 .....

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2 .....

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3 .....

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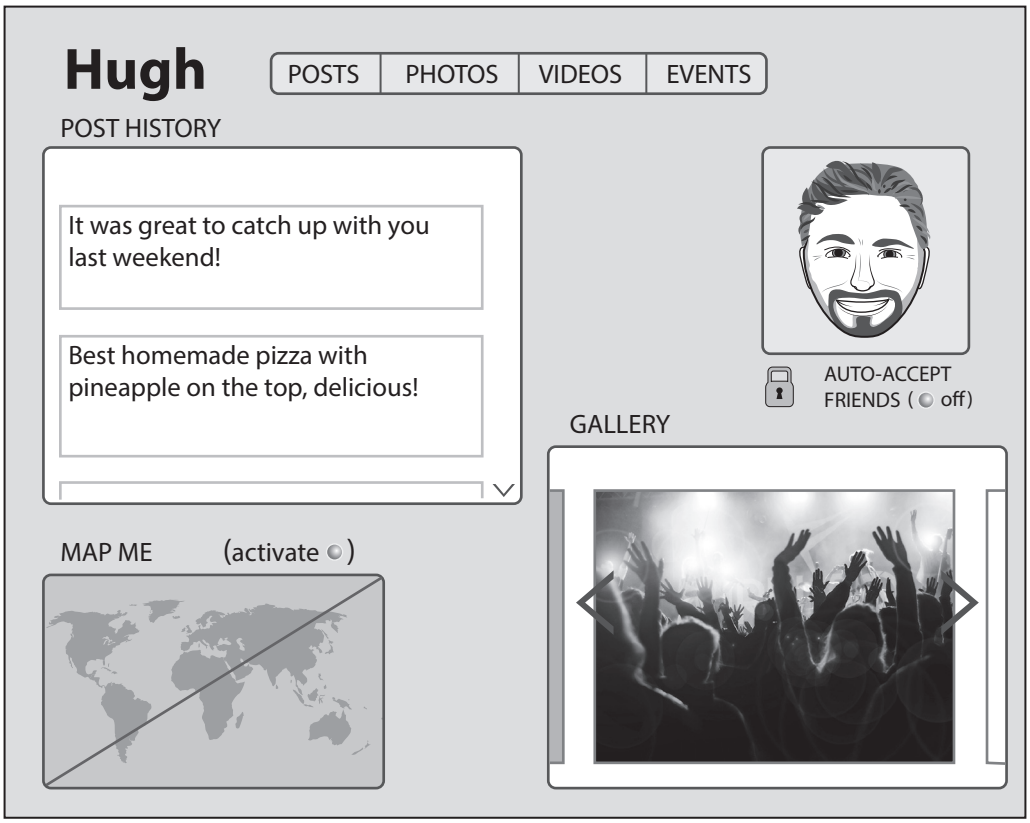
(c) **Figure 1** shows an image of Hugh's social media settings.



(Source: © PAL)

Figure 1

Explain **two** ways Hugh has protected his personal information.

(4)

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2 .....

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(d) Hugh is choosing the type of computer he will use to manage the purchase and delivery of audio files to customers.

Explain why an embedded system is a poor choice.

(2)

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(e) Hugh is working on a musical project.

(i) List **two** key features of sound editing software that Hugh could use to create a song.

(2)

1 .....

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2 .....

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(ii) Describe **two** key features of project management software that Hugh could use to make sure the project is complete before the deadline.

(4)

1 .....

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2 .....

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(f) Hugh uses a desktop computer to write, create and distribute his music.

Discuss the advantages and disadvantages to Hugh of selecting a desktop computer rather than a laptop, smartphone or tablet computer.

In your answer you should consider: components, portability, connectivity, cost and maintenance.

(8)

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(Total for Question 4 = 24 marks)

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5 Hugh creates a website for a band.

(a) Explain **one** drawback to the website users of Hugh reducing the file size of images used on the website.

(2)

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(b) Describe the function of **two** network components that are used to transfer data on the Internet.

(4)

Network component .....

Function .....

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Network component .....

Function .....

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(c) Explain why streaming allows users with limited storage on their devices to listen to the band's music.

(2)

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(d) Describe **one** method that should be used to secure payment data transferred between Hugh's server and other computers.

(3)

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(e) Describe how WPA is a more secure protocol than WEP.

(2)

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(f) Hosted applications have fewer features than local applications.

Explain **one** positive impact on the user experience of reducing the number of features.

(2)

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(g) Evaluate how online communities have impacted on how we socialise.

(8)

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(Total for Question 5 = 23 marks)

TOTAL FOR PAPER = 100 MARKS

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